

SYSTEM OVERVIEW :

BLOCK DIAGRAM :

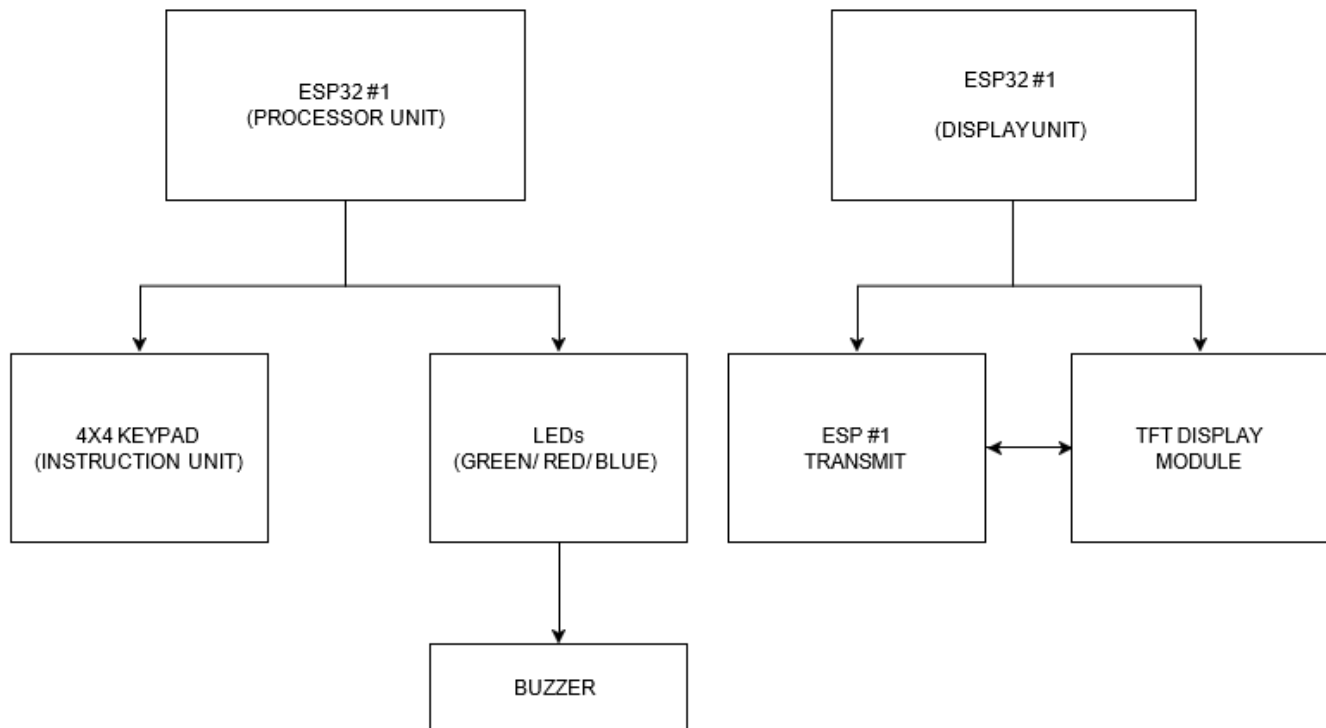


FIGURE 1

The objective of the Mini 8086 Microprocessor Simulation System is to use two ESP32 microcontrollers to simulate the basic functions of the 8086 microprocessor. The input unit (4x4 matrix keypad), the processing unit (ESP32 #1), and the output/display unit (ESP32 #2 with TFT Display Module) represent the three primary parts of the system.

An instruction input by the user using the keypad will be processed by ESP32 #1, which then sends the results to ESP32 #2 after modifying the register values appropriately. The simulated microprocessor's current state is then shown on a TFT Display Module using ESP32 #2 function.

SYSTEM OVERVIEW :

TFT Display Module

- The TFT Display Module is used to visually present the simulated 8086 microprocessor's register values, executed instructions, and operation results.

4x4 Matrix Keypad

- The 4x4 Matrix Keypad is used to input instructions, select operations, and assign registers for the simulated 8086 microprocessor.

ESP 32 #1

- The ESP 32 #1 functions as the core processor that simulates the 8086 microprocessor's operations, including handling input from the keypad, executing ALU instructions, and managing register data.

ESP 32 #2

- The ESP 32 #2 handles the TFT display by receiving data from ESP32 #1 and presenting the simulated 8086 microprocessor's instructions, register values, and operation results visually.

Buzzer

- The Buzzer alerts the user with a sound whenever an invalid input or error occurs during the simulated 8086 microprocessor operations.

LEDs

- The LEDs indicate whether you have reset, executed, or entered the registers.